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Clémence Prévost

Resume

Education

2018–now **Ph.D. in Signal Processing**, University of Lorraine, CNRS, CRAN, Nancy, France.

- **Ph.D. Thesis:** "*Multimodal data fusion via low-rank approximations*," under supervision of **David Brie** and **Konstantin Usevich**.

2017–2018 **M.Sc. in System Engineering**, *University of Lorraine, Nancy, France.*

- **M.Sc. Thesis:** "*Low-rank structured matrix completion in the context of MRI*," under supervision of **David Brie** and **Konstantin Usevich**.

2015–2018 **M.Sc. in Electrical and Mechanical Engineering**, *École Nationale Supérieure d'Electricité et de Mécanique (ENSEM), Nancy, France.*

2013–2015 **CPGE PCSI - PSI*, Lycée César Baggio, Lille.**, *Undergraduate studies to prepare for competitive entry exams to engineering schools.*

2013 **French Baccalaureate with honors, Lycée Denis Diderot, Langres, France**, *Biology, Physics, Mathematics (speciality in Linear algebra).*

Research Interests

- **Multimodal data fusion**;
- **Low-rank approximation**;
- **Tensor factorization**;
- **Cramér-Rao bounds**.

Publications

Journal publications

- "Hyperspectral super-resolution with coupled Tucker approximation: Identifiability and SVD-based algorithms", Prévost, C., Usevich, K., Brie, D. and Comon, P., in *IEEE Transactions on Signal Processing, vol. 68, p.931-946.*

International conferences

- "Cramér-Rao Lower Bounds with random equality constraints", Prévost, C., Chaumette, E., Usevich, K., Brie, D. and Comon, P., in *2020 IEEE ICASSP (International Conference on Acoustics, Speech and Signal Processing).*

- "Cramér-Rao Bounds in the framework of hyperspectral super-resolution", Prévost, C., Usevich, K., Haardt, M., Brie, D. and Comon, P., in *2019 IEEE CAMSAP (international workshop on Computational Advances in Multi-sensor Adaptative Processing)*.
- "Coupled tensor low-rank multilinear approximation for hyperspectral super-resolution", Prévost, C., Usevich, K., Brie, D. and Comon, P., in *2019 IEEE ICASSP (International Conference on Acoustics, Speech and Signal Processing)*.

National conferences

- "Approches tensorielles couplées pour la fusion aveugle d'images multispectrale et hyperspectrale", Prévost, C., Usevich, K., Brie, D. and Comon, P., in *2019 GRETSI (Colloque francophone de traitement de signal et des images)*.

Workshops and seminars

- **Hyperspectral super-resolution via coupled Tucker decomposition**. Workshop on Low-Rank Models and Applications (LRMA), September 2019.
- **Hyperspectral super-resolution via coupled Tucker decomposition**. Peyresq Summer school, July 2019.
- **Hyperspectral super-resolution via coupled Tucker decomposition**. GdR ISIS meeting on tensor decompositions, June 2019.

Teaching Experiences

2018–2021 **Teaching assistant**, IUT Nancy-Brabois, Department of Telecommunications and Networks (R&T), Nancy, France.

- **Practical courses**: study of first and second order filters, introduction to frequency analysis, frequency modulation (192h).

Activities

- ENBIS-18, Nancy, France: registration desk, technical chair;
- IEEE CAMSAP-19, Le Gosier, Guadeloupe: registration desk.

Skills

- **Programming skills**: MATLAB and Simulink, Python, R.

Languages

- **French:** Fluent;
- **English:** Fluent (TOEIC: 950/990).